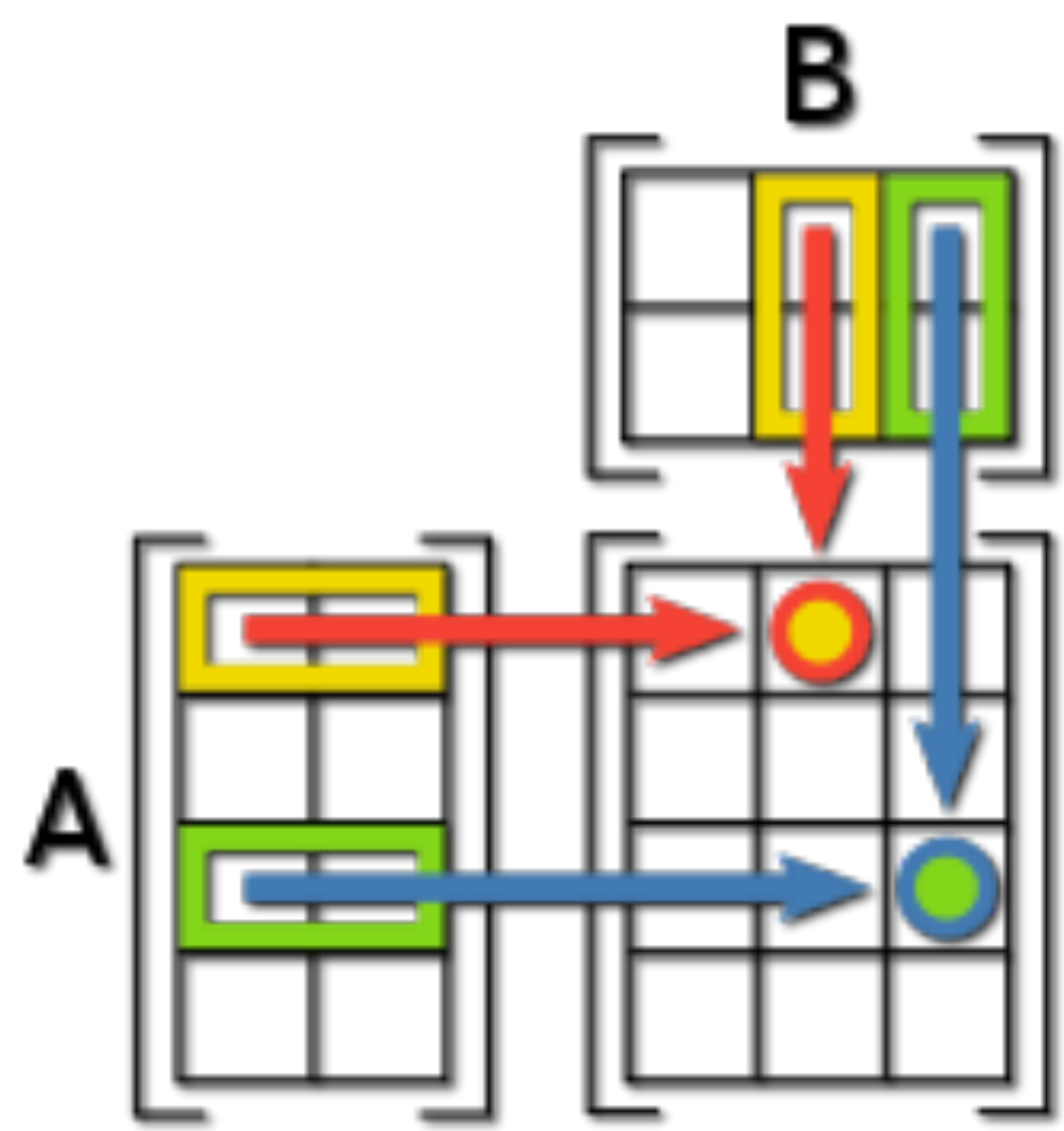


# Matrices

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# Matrices

- A matrix is a two-dimensional array of numbers

- ▶ having  $M$  rows and  $N$  columns

- matrix  $\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1N} \\ & & \vdots & \\ a_{M1} & a_{M2} & \dots & a_{MN} \end{pmatrix}$  commonly stored as 2D array  $A = \begin{bmatrix} [a_{11}, a_{12}, \dots, a_{1N}], \dots, [a_{M1}, \dots, a_{MN}] \end{bmatrix}$

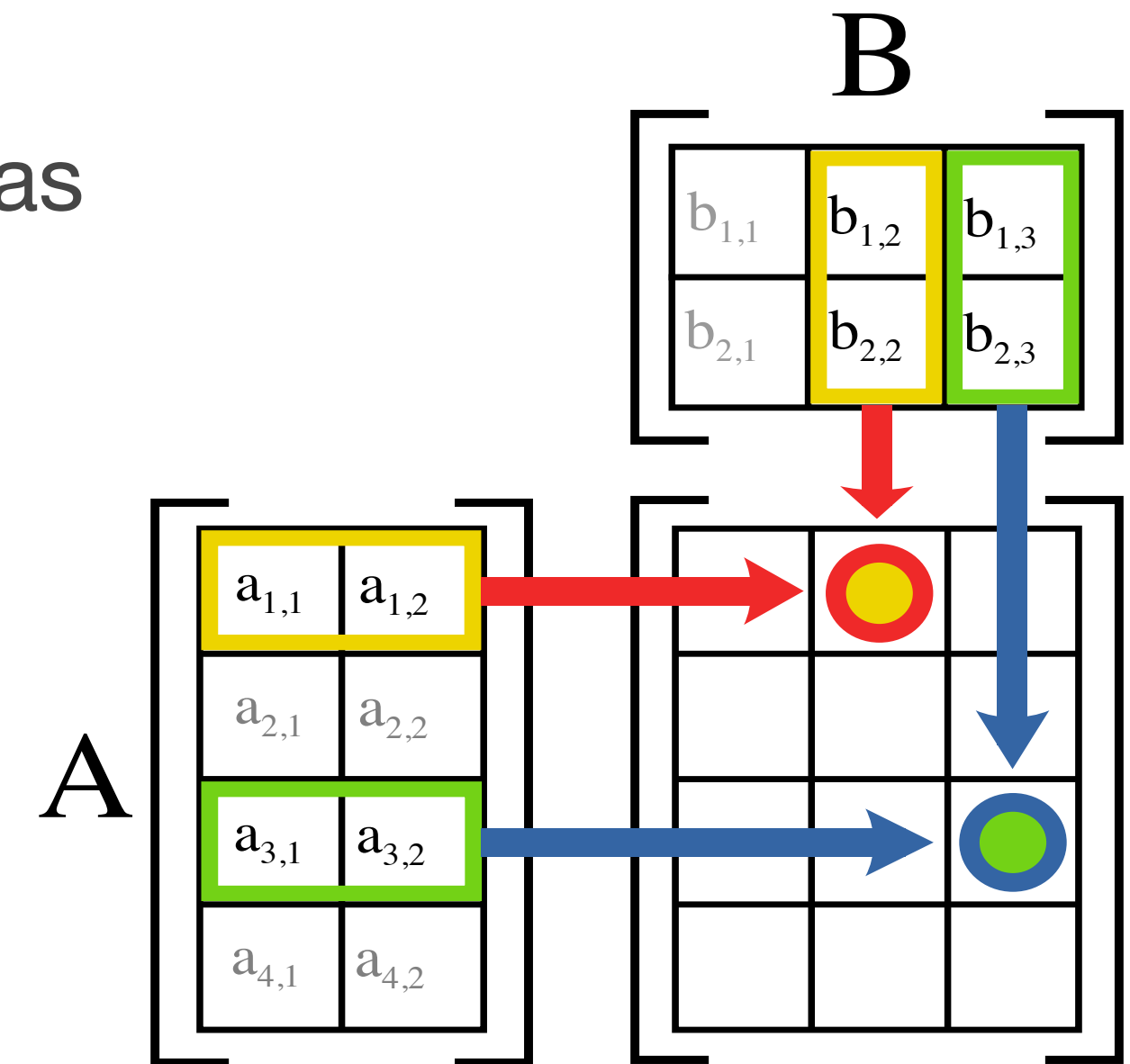
- Two matrices  $\mathbf{A}$  and  $\mathbf{B}$  are equal iff  $a_{ij} = b_{ij}$  for all  $i, j$
- The sum of two matrices  $\mathbf{A}$  and  $\mathbf{B}$  is  $c_{ij} = a_{ij} + b_{ij}$ 
  - ▶ analogous for the subtraction/difference
- The scalar multiple  $\mathbf{C} = s \cdot \mathbf{A}$  is  $c_{ij} = s \cdot a_{ij}$

# Matrix Multiplication and Inverse

- For an  $M \times N$  matrix  $\mathbf{A}$  and  $N \times P$  matrix  $\mathbf{B}$  we get an  $M \times P$  matrix  $\mathbf{C} = \mathbf{AB}$  as

$$c_{ij} = \sum_{s=1}^M a_{is} \cdot b_{sj}$$

- ▶ element  $c_{ij}$  of  $\mathbf{C}$  is dot-product of  $j$ -th column vector from  $\mathbf{B}$  with the  $i$ -th row vector from  $\mathbf{A}$ 
  - matrix multiplication is not commutative  $\mathbf{AB} \neq \mathbf{BA}$
  - matrix multiplication is distributive  $\mathbf{A}(\mathbf{B}+\mathbf{C}) = \mathbf{AB} + \mathbf{AC}$
- The inverse  $\mathbf{A}^{-1}$  has the property that
$$\mathbf{A}^{-1} \mathbf{A} = \mathbf{A} \mathbf{A}^{-1} = \mathbf{I}$$
  - ▶  $\mathbf{I}$  being the identity matrix
  - ▶ inverse only exists for square matrices with non-zero determinant



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# Matrix-Vector Multiplication

- Matrix-vector multiplication  $\mathbf{A}\mathbf{v}$  corresponds to matrix product rule

$$\mathbf{u} = \mathbf{A} \cdot \mathbf{v} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1N} \\ & & \vdots & \\ a_{M1} & a_{M2} & \dots & a_{MN} \end{pmatrix} \cdot \begin{pmatrix} v_1 \\ \vdots \\ v_N \end{pmatrix}$$

- Column vector  $\mathbf{v}$  is multiplied (as dot-product) with each row of  $\mathbf{A}$ 
  - ▶ number of columns of  $\mathbf{A}$  must match rows of  $\mathbf{v}$
- Result vector  $\mathbf{u}$  has  $M$  entries (not  $N$ )
  - ▶ same as number of rows of  $\mathbf{A}$

# Transpose

- Transposition  $\mathbf{B} = \mathbf{A}^T$  is achieved by mirroring entries about the upper-left to lower-right diagonal

- ▶ row/column indices and dimensions are swapped  $b_{ij} = a_{ji}$

- Matrix  $\mathbf{A}$  is said to be symmetric if  $\mathbf{A} = \mathbf{A}^T$

- An  $N$ -dimensional vector  $\mathbf{a}$  can be considered an  $N \times 1$  matrix

- ▶ also called a column vector  $\mathbf{a} = \begin{pmatrix} a_1 \\ \vdots \\ a_N \end{pmatrix}$

- ▶ its transpose is then a row vector  $\mathbf{a}^T = (a_1 \ a_2 \ \dots \ a_N)$

- Convenient matrix notation of dot-product between vectors  $\mathbf{a}$  and  $\mathbf{b}$

$$\mathbf{a} \circ \mathbf{b} = \mathbf{a}^T \cdot \mathbf{b} = (a_1 \ \dots \ a_N) \cdot \begin{pmatrix} b_1 \\ \vdots \\ b_N \end{pmatrix}$$

**A**

|   |   |
|---|---|
| 1 | 2 |
| 3 | 4 |
| 5 | 6 |

# Matrices in Graphics

- Typically used in graphics for operations on (position) vectors
  - ▶ in computer graphics most matrices are typically of dimensions 2x2, 3x3 or 4x4
- Vertex coordinates and normal vectors correspond to 3-dimensional vectors  $\mathbf{v}^T = [x, y, z]$ 
  - ▶ column vectors, equivalent to a 3x1 matrix
- Matrix-vector products defining linear transformations on its operands
  - ▶ used for geometric transformations and projections
- Recall dot-product being:  $\mathbf{u}^T \cdot \mathbf{v}$
- Outer product:  $\mathbf{u} \cdot \mathbf{v}^T$  of two vectors forms a new matrix

$$\mathbf{u} \cdot \mathbf{v}^T = \begin{pmatrix} u_1 \\ \vdots \\ u_N \end{pmatrix} \cdot (v_1 \ \dots \ v_N) = \begin{pmatrix} u_1 v_1 & u_1 v_2 & \dots & u_1 v_N \\ & \vdots & & \\ u_M v_1 & u_M v_2 & \dots & u_M v_N \end{pmatrix}$$



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